

## FINAL EXAM LEARNING GUIDE

1. Listed below are the Learning Objectives for Units 4-6. In the table below write your own exam question that covers the given objective. Use this to structure your own learning guide and better focus your study time. Also determine your confidence with each objective with the numerical key below.

**KEY: 1= Not Confident      2 = Somewhat Confident**  
**3 = Absolutely Confident**

| Unit Learning Objective                                                                   | Create Your Own Questions | Confidence | Prior Exam Performance |
|-------------------------------------------------------------------------------------------|---------------------------|------------|------------------------|
| Unit 4A                                                                                   |                           |            |                        |
| Use basic trigonometry to solve right triangles.                                          |                           |            |                        |
| Understand and use the reciprocal, quotient, and Pythagorean identities.                  |                           |            |                        |
| Construct the Unit Circle using the relationships found in special right triangles.       |                           |            |                        |
| Know the sign of the trigonometric values in each quadrant.                               |                           |            |                        |
| Use trigonometric identities to simplify trigonometric expressions.                       |                           |            |                        |
| Solve trigonometric equations by using trigonometric identities and algebraic techniques. |                           |            |                        |
| Calculate trigonometric values from angles found on the unit circle.                      |                           |            |                        |
| Find trigonometric ratios based on a given trigonometric value in a specified quadrant.   |                           |            |                        |
| Graph the parent trigonometric functions: sine, cosine, tangent.                          |                           |            |                        |

| Unit Learning Objective                                                                                     | Create Your Own Questions | Confidence | Prior Exam Performance |
|-------------------------------------------------------------------------------------------------------------|---------------------------|------------|------------------------|
| <b>Unit 4B</b>                                                                                              |                           |            |                        |
| Graph and determine transformations of trigonometric functions from an equation, including key points.      |                           |            |                        |
| Know the domain and range of inverse trigonometric functions to evaluate inverse trig expressions.          |                           |            |                        |
| Use knowledge of inverse trigonometric functions to simplify composite trig expressions.                    |                           |            |                        |
| Write an equation for a sinusoidal function given key points from the graph.                                |                           |            |                        |
| <b>Unit 5</b>                                                                                               |                           |            |                        |
| Identify the intercepts, end behavior, and discontinuities of rational functions.                           |                           |            |                        |
| Use attributes and graphs of rational functions to write the equation.                                      |                           |            |                        |
| Determine the limit of a function using graphs and tables.                                                  |                           |            |                        |
| Apply the power rule to differentiate equations.                                                            |                           |            |                        |
| Evaluate the instantaneous rate of change of a function at a given x-value.                                 |                           |            |                        |
| Use derivatives to determine the increasing and decreasing intervals and the relative extrema of functions. |                           |            |                        |
| <b>Unit 6</b>                                                                                               |                           |            |                        |
| Graph parametric equations.                                                                                 |                           |            |                        |

| Unit Learning Objective                                                    | Create Your Own Questions | Confidence | Prior Exam Performance |
|----------------------------------------------------------------------------|---------------------------|------------|------------------------|
| Convert between rectangular relations and parametric equations.            |                           |            |                        |
| Create parametric equations to model real-world problems.                  |                           |            |                        |
| Determine the component form and magnitude given a vector on a plane.      |                           |            |                        |
| Determine the magnitude and direction of a vector given in component form. |                           |            |                        |
| Graph and label given points in the polar coordinate system.               |                           |            |                        |
| Convert between rectangular and polar coordinates.                         |                           |            |                        |
| Graph polar equations.                                                     |                           |            |                        |
| Write equations of polar functions given a graph.                          |                           |            |                        |

2. What are your goals for the Final Exam? Remember that a Satisfaction Goal is the grade you would need to be satisfied with your work, and a Pride Goal is the grade necessary for you to feel pride in your work.

Satisfaction Goal = \_\_\_\_\_/100

Pride Goal = \_\_\_\_\_/100